7 **(a)** A sample of a radioactive isotope contains N nuclei of the isotope at time T. At time $(T + \Delta T)$, the sample contains $(N - \Delta N)$ nuclei of the isotope. The time interval ΔT is short.

Use the symbols N, ΔN, T and ΔT to give expressions for:

(i) the average activity of the sample during the time ΔT

..... [1]

(ii) the probability of decay of a nucleus in the time ΔT

..... [1]

(iii) the decay constant λ of the isotope.

..... [1]

(b) The isotope polonium-208 ($^{208}_{84}\text{Po}$) is radioactive and decays to form lead-204 ($^{204}_{82}\text{Pb}$). The nuclear equation for this decay is

$$^{208}_{84}\text{Po} \rightarrow {}^{204}_{82}\text{Pb} + {}^{4}_{2}\text{He}$$

Data for the nuclei in the reaction are given in Table 7.1.

Table 7.1

nucleus	mass / u
$^{4}_{2}\text{He}$	4.002603
$^{204}_{82}\text{Pb}$	203.973043
$^{208}_{84}\text{Po}$	207.981245

(i) Determine, to three decimal places, the energy released, in MeV, associated with the decay of polonium-208.

energy = ... MeV [3]

(ii) The polonium-208 nucleus is initially stationary. The initial kinetic energy of the $^{4}_{2}He$ nucleus (α-particle) is found to be less than the energy calculated in **7b(i)**.

Suggest **two** possible reasons for this difference.

1. .....

.....

2. .....

.....

[2]

(c) Lanthanium-140 ($^{140}_{57}La$) emits a β-particle to become a stable nuclide, cerium (Ce). The decay is represented by the equation

$$^{140}_{57}La \rightarrow \beta + Ce$$

(i) State the nuclear notation for cerium.

$$^{\cdots\cdots}_{\cdots\cdots}Ce$$

[1]

(ii) Lanthanum-140 has a half-life of $T_{1/2}$.

A pure sample of lanthanum-140 contains N_0 nuclei.

The variation with time t of the number N of lanthanum-140 nuclei is shown in Figure 7.1.

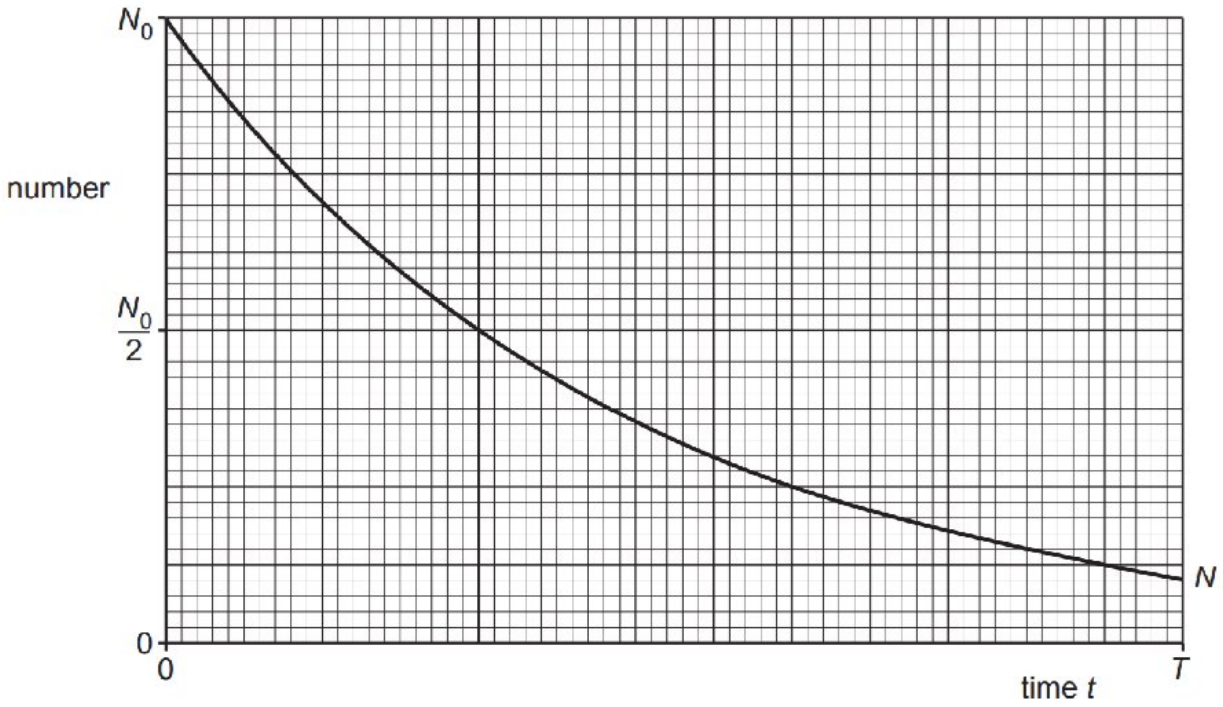


Figure 7.1

On Figure 7.1:

- label, on the time axis, the time $t = 1.0T_{1/2}$ and the time $t = 2.0T_{1/2}$.
- sketch the variation with time t of the number of nuclei of cerium for $t = 0$ to $t = T$.

[2]

(d) In 2024, scientists in Montana discovered a fossil of the dinosaur, *Lokiceratops*.

To find the age of the fossil, scientists measured a radioactive isotope in a mineral sample taken from the volcanic ash layer surrounding the fossil.

The half-life of the isotope is 7.80×10^7 years.

A 12 g sample of new mineral sample from fresh volcanic rock has an activity of 36 Bq.

A 4 g sample of ancient mineral from the fossil's ash layer has an activity of 5.0 Bq.

Calculate the age of the fossil.

age = ... years [2]

[Total: 13]